

LAB 8

Business Idea Management System

Objective

Develop a backend application for managing **business ideas**, allowing users to register, log in/out, and manage their submitted ideas.

The system includes CRUD operations, JWT authentication, and Mongoose populate() to display related data.

System Overview

The backend manages three main collections in MongoDB:

- **Users** – account and authentication information
- **Ideas** – business idea submissions
- **Evaluations** – review and scoring of each idea

1. Users Collection

Field	Type	Description
_id	ObjectId	Unique identifier
username	String	Must be unique
password	String	Encrypted using bcrypt
role	String	Default: "user" (optional: "admin")
createdAt	Date	Timestamp when registered

Example:

```
{
  "_id": "674b2a90b82af93a2b17a0f1",
  "username": "minh_startup",
  "password": "$2b$10$hash...",
  "role": "user",
  "createdAt": "2025-11-03T09:00:00Z"
}
```

2. Ideas Collection

Field	Type	Description
_id	ObjectId	Unique identifier
idea_title	String	Title of the business idea
description	String	Detailed explanation of the idea
potential_score	Number	Evaluation score (1–10)
created_at	Date	Timestamp when idea was created

Example:

```
{
  "_id": "674b3a11a82a0d6f12345678",
  "idea_title": "Eco-Friendly Coffee Cups",
  "description": "Biodegradable coffee cups made from rice husks.",
  "potential_score": 8.5,
  "created_at": "2025-11-03T09:15:00Z"
}
```

3. Evaluations Collection

Field	Type	Description
-------	------	-------------

_id	ObjectId	Unique identifier
userId	ObjectId	Ref → Users
ideaId	ObjectId	Ref → Ideas
score	Number	Reviewer's score (1–10)
feedback	String	Reviewer's comment
created_at	Date	Date/time of evaluation

Relationships

- A **User** can submit multiple **Ideas**
- A **User** can also evaluate others' ideas
- An **Idea** can receive multiple evaluations

General Requirements

- **Database:** BusinessIdeaDB_2025
- **Folder name:** IdeaSys_StudentCode
- **Tools:** Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, dotenv, EJS

A `/resources` folder is provided with:

- Sample JSON data
- List of required npm packages
- Example EJS template for rendering an idea detail page
- Postman collection for API testing

Tasks to be Completed

Task 1 – Project Initialization & Configuration

Create a Node.js project named `IdeaSys_StudentCode`.

Install dependencies:

```
npm install express mongoose bcrypt jsonwebtoken dotenv cors ejs
```

Create `.env` file:

```
PORT=1234
MONGO_URI=your_mongodb_connection
JWT_SECRET=mysecretkey
```

Set up MongoDB database `BusinessIdeaDB_2025` with three collections: Users, Ideas, Evaluations.

Task 2 – User Authentication APIs

2.1 POST /users/register

Registers a new user with `username`, `password`.
Password must be hashed.

2.2 POST /users/login

Validates credentials, returns JWT token.

2.3 POST /users/logout

Simulate logout and return JSON confirmation.

Task 3 – Ideas Management (CRUD APIs)

Method	Endpoint	Description	Auth
GET	/ideas	Get all business ideas	Public
GET	/ideas/:id	Get details by ID	Public
POST	/ideas	Submit new idea	Auth (user)
PUT	/ideas/:id	Update idea	Auth (owner/admin)
DELETE	/ideas/:id	Delete idea	Auth (owner/admin)

Example create request:

```
{
  "idea_title": "AI-Powered Tutor App",
  "description": "A mobile app that uses AI to provide personalized learning paths.",
  "potential_score": 9
}
```

Task 4 – Evaluations (Authenticated Users)

Method	Endpoint	Description
POST	/evaluations/:ideaId	Add a new evaluation
GET	/evaluations/myreviews	View all evaluations by logged-in user
PUT	/evaluations/:id	Update evaluation
DELETE	/evaluations/:id	Delete evaluation

When a new evaluation is added, **recalculate the idea's average potential_score**.

Use `.populate('ideaId', 'idea_title')` to show related idea info.

Example GET response:

```
{
  "user": "minh_startup",
  "evaluations": [
    {
      "idea": { "idea_title": "Eco-Friendly Coffee Cups" },
      "score": 9,
      "feedback": "Great market potential!",
      "created_at": "2025-11-03T09:30:00Z"
    }
  ]
}
```

Task 5 – Advanced Data Display Using populate()

5.1 GET /ideas/details/:id (Authenticated)

Returns full details of an idea, including evaluations and reviewer usernames.

Use nested populate.

Example:

```
{
  "idea_title": "Smart Recycling Bins",
```

```

    "description": "IoT-enabled bins that reward users for recycling.",
    "potential_score": 8.7,
    "evaluations": [
      {
        "score": 9,
        "feedback": "Innovative concept with social impact.",
        "user": { "username": "huy_innovator" }
      }
    ]
  }
}

```

Task 6 –Optional: Render Handlebars View (hbs)

Create a Handlebars view to render `/ideas/details/:id` data.

6.1. Setup View Engine

In your `server.js`:

```

import express from "express";
import hbs from "hbs";
import path from "path";

const app = express();
app.set("view engine", "hbs");
app.set("views", path.join(process.cwd(), "views"));

```

6.2. Create View File: `views/idea-details.hbs`

```

<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>{{idea_title}}</title>
    <style>
      body { font-family: Arial; margin: 40px; }
      h1 { color: #2c3e50; }
      .idea-card { border: 1px solid #ccc; border-radius: 10px; padding: 20px; }
    </style>
  </head>
  <body>
    <div class="idea-card">
      <h1>{{idea_title}}</h1>
      <p><strong>Description:</strong> {{description}}</p>
      <p><strong>Average Potential Score:</strong> {{potential_score}}</p>
    </div>

    <div class="reviews">
      <h2>Evaluations</h2>
      {{#if evaluations.length}}
        {{#each evaluations}}
          <div class="review">
            <p><strong>User:</strong> {{user.username}}</p>
            <p><strong>Score:</strong> {{score}}</p>
            <p><strong>Feedback:</strong> {{feedback}}</p>
          </div>
        {{/each}}
      {{else}}
        <p>No evaluations yet.</p>
      {{/if}}
    </div>
  </body>
</html>

```

```
    </div>
  </body>
</html>
```

6.3. API Endpoint Example

```
app.get("/ideas/details/:id", async (req, res) => {
  const idea = await Idea.findById(req.params.id)
    .populate({
      path: "evaluations",
      populate: { path: "userId", select: "username" },
    });

  res.render("idea-details", idea);
});
```

Expected Learning Outcomes

After completing this lab, students will:

- Build full CRUD APIs with Express + MongoDB
- Implement JWT authentication and authorization
- Use Handlebars for dynamic rendering
- Apply `populate()` for relationships in MongoDB

Deliverables

- Node.js project folder: `IdeaSys_StudentCode`
- `.env` configuration
- Postman collection of all APIs
- MongoDB Compass screenshots of collections
- Optional: EJS rendered page screenshot

JSON DATA

1. users.json

```
[
  {
    "_id": "67500001a1f9a1c001111111",
    "username": "minh_startup",
    "password": "$2b$10$hashedpass1",
    "role": "user",
    "createdAt": "2025-11-03T08:00:00Z"
  },
  {
    "_id": "67500001a1f9a1c002222222",
    "username": "lan_innovator",
    "password": "$2b$10$hashedpass2",
    "role": "user",
    "createdAt": "2025-11-03T08:10:00Z"
  },
  {
    "_id": "67500001a1f9a1c003333333",
    "username": "tuan_eco",
    "password": "$2b$10$hashedpass3",
    "role": "user",
    "createdAt": "2025-11-03T08:20:00Z"
  },
  {
    "_id": "67500001a1f9a1c004444444",
    "username": "mai_foodie",
    "password": "$2b$10$hashedpass4",
    "role": "user",
    "createdAt": "2025-11-03T08:30:00Z"
  },
  {
    "_id": "67500001a1f9a1c005555555",
    "username": "phuc_tech",
    "password": "$2b$10$hashedpass5",
    "role": "user",
    "createdAt": "2025-11-03T08:40:00Z"
  },
  {
    "_id": "67500001a1f9a1c006666666",
    "username": "thu_green",
    "password": "$2b$10$hashedpass6",
    "role": "user",
    "createdAt": "2025-11-03T08:50:00Z"
  },
  {
    "_id": "67500001a1f9a1c007777777",
    "username": "dat_ai",
    "password": "$2b$10$hashedpass7",
    "role": "user",
    "createdAt": "2025-11-03T09:00:00Z"
  },
  {
    "_id": "67500001a1f9a1c008888888",
    "username": "anh_travel",
    "password": "$2b$10$hashedpass8",
    "role": "user",
    "createdAt": "2025-11-03T09:10:00Z"
  },
  {
    "_id": "67500001a1f9a1c009999999",
    "username": "admin_idea",
    "password": "$2b$10$hashedpass9",
    "role": "admin",
    "createdAt": "2025-11-03T09:20:00Z"
  }
]
```

```

    },
    {
      "_id": "67500001a1f9a1c00aaaaaaa",
      "username": "khanh_health",
      "password": "$2b$10$hashedpass10",
      "role": "user",
      "createdAt": "2025-11-03T09:30:00Z"
    }
  ]

```

2. ideas.json

```

[
  {
    "_id": "67501001b2f9a2c001111111",
    "idea_title": "Eco-Friendly Coffee Cups",
    "description": "Biodegradable coffee cups made from rice husks to reduce plastic waste.",
    "potential_score": 8.7,
    "created_at": "2025-11-03T10:00:00Z"
  },
  {
    "_id": "67501001b2f9a2c002222222",
    "idea_title": "AI-Powered Tutor App",
    "description": "A personalized learning assistant app using AI to adapt to student needs.",
    "potential_score": 9.2,
    "created_at": "2025-11-03T10:05:00Z"
  },
  {
    "_id": "67501001b2f9a2c003333333",
    "idea_title": "Smart Waste Bins",
    "description": "IoT-enabled bins that automatically sort recyclable materials.",
    "potential_score": 8.9,
    "created_at": "2025-11-03T10:10:00Z"
  },
  {
    "_id": "67501001b2f9a2c004444444",
    "idea_title": "Plant-Based Fast Food Chain",
    "description": "Healthy fast food using 100% plant-based ingredients for urban consumers.",
    "potential_score": 9.0,
    "created_at": "2025-11-03T10:15:00Z"
  },
  {
    "_id": "67501001b2f9a2c005555555",
    "idea_title": "TravelMate App",
    "description": "AI-based travel planning app that builds itineraries based on interests.",
    "potential_score": 8.5,
    "created_at": "2025-11-03T10:20:00Z"
  },
  {
    "_id": "67501001b2f9a2c006666666",
    "idea_title": "Smart Hydroponic Garden",
    "description": "Indoor gardening system controlled via smartphone app.",
    "potential_score": 9.1,
    "created_at": "2025-11-03T10:25:00Z"
  },
  {
    "_id": "67501001b2f9a2c007777777",
    "idea_title": "HealthTrack Band",
    "description": "Affordable fitness band with real-time health monitoring for seniors.",
    "potential_score": 8.8,
    "created_at": "2025-11-03T10:30:00Z"
  }
]

```

```

    },
    {
      "_id": "67501001b2f9a2c008888888",
      "idea_title": "Sustainable Fashion Marketplace",
      "description": "Platform connecting eco-conscious designers and
customers.",
      "potential_score": 9.3,
      "created_at": "2025-11-03T10:35:00Z"
    },
    {
      "_id": "67501001b2f9a2c009999999",
      "idea_title": "ElderCare Robot",
      "description": "AI companion robot assisting elderly people with daily
activities.",
      "potential_score": 9.4,
      "created_at": "2025-11-03T10:40:00Z"
    },
    {
      "_id": "67501001b2f9a2c00aaaaaaa",
      "idea_title": "Online Farmers Market",
      "description": "E-commerce platform connecting local farmers directly to
consumers.",
      "potential_score": 8.6,
      "created_at": "2025-11-03T10:45:00Z"
    }
  ]
}

```

3 evaluations.json

```

[
  {
    "_id": "67502001c3f9a3c001111111",
    "userId": "67500001a1f9a1c001111111",
    "ideaId": "67501001b2f9a2c001111111",
    "score": 9,
    "feedback": "Excellent eco concept, strong market potential.",
    "created_at": "2025-11-03T11:00:00Z"
  },
  {
    "_id": "67502001c3f9a3c002222222",
    "userId": "67500001a1f9a1c002222222",
    "ideaId": "67501001b2f9a2c002222222",
    "score": 10,
    "feedback": "High scalability with strong demand in edtech.",
    "created_at": "2025-11-03T11:05:00Z"
  },
  {
    "_id": "67502001c3f9a3c003333333",
    "userId": "67500001a1f9a1c003333333",
    "ideaId": "67501001b2f9a2c003333333",
    "score": 8,
    "feedback": "Smart waste solution but needs government cooperation.",
    "created_at": "2025-11-03T11:10:00Z"
  },
  {
    "_id": "67502001c3f9a3c004444444",
    "userId": "67500001a1f9a1c004444444",
    "ideaId": "67501001b2f9a2c004444444",
    "score": 9,
    "feedback": "Vegan food trend fits perfectly with urban lifestyle.",
    "created_at": "2025-11-03T11:15:00Z"
  },
  {
    "_id": "67502001c3f9a3c005555555",
    "userId": "67500001a1f9a1c005555555",
    "ideaId": "67501001b2f9a2c005555555",
    "score": 8,

```

```

    "feedback": "Travel AI concept is interesting, UI could improve.",
    "created_at": "2025-11-03T11:20:00Z"
  },
  {
    "_id": "67502001c3f9a3c006666666",
    "userId": "67500001a1f9a1c006666666",
    "ideaId": "67501001b2f9a2c006666666",
    "score": 9,
    "feedback": "Strong innovation in agri-tech, great sustainability
impact.",
    "created_at": "2025-11-03T11:25:00Z"
  },
  {
    "_id": "67502001c3f9a3c007777777",
    "userId": "67500001a1f9a1c007777777",
    "ideaId": "67501001b2f9a2c007777777",
    "score": 9,
    "feedback": "Good for elderly care, price point could be lower.",
    "created_at": "2025-11-03T11:30:00Z"
  },
  {
    "_id": "67502001c3f9a3c008888888",
    "userId": "67500001a1f9a1c008888888",
    "ideaId": "67501001b2f9a2c008888888",
    "score": 10,
    "feedback": "Sustainable fashion is booming; perfect timing.",
    "created_at": "2025-11-03T11:35:00Z"
  },
  {
    "_id": "67502001c3f9a3c009999999",
    "userId": "67500001a1f9a1c009999999",
    "ideaId": "67501001b2f9a2c009999999",
    "score": 10,
    "feedback": "AI robotics will dominate the healthcare sector.",
    "created_at": "2025-11-03T11:40:00Z"
  },
  {
    "_id": "67502001c3f9a3c00aaaaaaaa",
    "userId": "67500001a1f9a1c00aaaaaaaa",
    "ideaId": "67501001b2f9a2c00aaaaaaaa",
    "score": 9,
    "feedback": "Great idea supporting local farmers, social impact strong.",
    "created_at": "2025-11-03T11:45:00Z"
  }
]

```

✓ Tổng kết

File	Số bản ghi	Mối quan hệ
users.json	10	User có thể đánh giá nhiều idea
ideas.json	10	Mỗi idea có thể có nhiều evaluations
evaluations.json	10	Kết nối userId ↔ ideaId chuẩn 24 ký tự ObjectId